

Lydia Chien

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EDUCATION

National Yang Ming Chiao Tung University

Sep 2022 – May 2027

Bachelor of Electrical Engineering (GPA: 4.16/4.3)

Hsinchu, Taiwan

- **Coursework:** Control Systems, Robotics, Digital Signal Processing, Embedded Systems, Machine Learning
- **Research:** NYCU ICCL Lab (Multi-UAV Localization), NYCU Systems Engineering Lab (Autonomous Vehicles), Vision-Tactile Policy Learning

National University of Singapore (NUS)

Aug 2026 – Dec 2026

Exchange Student

Singapore

EXPERIENCE

Industrial Technology Research Institute (ITRI)

Feb 2026 – Jul 2026

Robotics Research Intern

Hsinchu, Taiwan

Project 1: Autonomous Warehouse Picking System

- Engineered a full robotic pick-and-place pipeline from scratch on an Elephant Robotics Pro 630 6-DOF arm (ROS 2 Humble / MoveIt 2), achieving 70% end-to-end success rate with a complete object-detection-to-placement cycle under 100 seconds
- Built a vision-guided suction detection system using SAM2 segmentation on a dual camera setup (world-mounted + wrist-mounted), integrating hand-eye calibration and ArUco marker tracking to dynamically add and remove collision objects in MoveIt's live octomap planning scene.

Project 2: VR Teleoperation for Imitation Learning Data Collection

- Built a bilateral teleoperation interface using Meta Quest 2 VR controllers to let an operator intuitively drive a 7-DOF dual-arm robot in real time — capturing high-quality human demonstrations for downstream imitation learning
- Engineered TRAC-IK solver with workspace retargeting and EMA smoothing, achieving **97% IK success rate**; implemented Swing-Twist quaternion decomposition for full 3-axis end-effector orientation tracking

PROJECTS

Vision-Tactile Manipulation Policy (Independent Research)

- Developing a contact-aware manipulation policy using UMI hand-held demonstrations with synchronized GoPro + DIGIT tactile sensor data; designed clapperboard-analog sync pipeline via pressure-spike detection and Δt estimation
- Proposed cross-attention fusion integrating SparSH tactile tokens with visual and language tokens into a slow-fast Reactive Diffusion Policy (RDP); evaluating on fragile objects (egg, tofu, test tube) with ablation: w/ vs. w/o tactile

Multi-UAV Indoor Localization System (UWB)

- Implemented Two-Way Ranging (TWR) on ESP32-controlled UWB modules; applied multilateration and least-squares optimization in Python, achieving **<10 cm accuracy** in a 20 m × 20 m space
- Competed at Singapore Amazing Flying Machine Competition (SAFMC 2025), placing **4th** in the Multi-UAV Collaboration category

LEADERSHIP & VOLUNTEERING

President, NYCU Makereal Labs — led 30-member team, hosted workshops for 100+ students

Volunteer, TWiST (Taiwan Women in Science & Tech) — mentored 80+ female high school students

HONORS & AWARDS

CTCI Foundation Science and Technology Scholarship (Creativity Award) — Autonomous UAV System

Winner, ADI Signal Measurement Competition

SKILLS

Programming : Python, C/C++, PyTorch | ROS 2, MoveIt 2, Gazebo

Robotics : Intel RealSense, DIGIT Tactile Sensor, SAM2, Octomap, TRAC-IK, UWB | Linux, Git, Docker